Thermodynamic Power Cycles

Complete Unabridged Master Notes (Pages 1 to 38)

1. Introduction to Gas Power Cycles & Air-Standard Assumptions

Gas Power Cycles

Gas power cycles form the basis of power plant systems where the working fluid is a gas (normally air, or products of air-fuel combustion). These cycles include internal combustion engine cycles (***Otto, Diesel, Dual/Mixed***), ***Ericsson***, and ***Stirling*** cycles.

Internal Combustion Engine Cycles

One of the most important power-producing devices is the internal combustion engine, so called because the combustion process occurs internally within the engine and not outside it as with external combustion engines like the Rankine cycle. Power plants are classified into gas turbine engines and reciprocating engines. Examples of reciprocating IC engines are Diesel engines (Rudolf Diesel) and Petrol engines (Nikolaus A. Otto, 1876).

Air-Standard Cycles

Both Otto's and Diesel's engines operate on open cycles where fresh air must be drawn in. To analyze them theoretically on a thermodynamic cycle, **air-standard engines** assume:

1. The working fluid is air.
2. Heat is added from an external source rather than fuel burning.
3. A heat sink is provided for heat rejection to return air to its original state.
4. All processes are internally reversible.
5. Air has constant specific heats evaluated at room temperature ($c_p = 1.005$ kJ/kg·K, $c_v = 0.718$ kJ/kg·K, $\gamma = 1.4$).

2. Air-Standard Otto Cycle

Ideal cycle for spark-ignition engines consisting of 4 processes: Process 1→2 (Isentropic compression), Process 2→3 (Constant-volume heat addition), Process 3→4 (Isentropic expansion), Process 4→1 (Constant-volume heat rejection).

Compression Ratio: $r_v = V_1 / V_2 = (V_s + V_c) / V_c$

Efficiency: $\eta_{th,\ Otto} = 1 - (1 / r_v^{\gamma-1})$

EXAMPLE 1 — OTTO CYCLE EFFICIENCY

Problem: Petrol engine with bore $B = 58$ mm, stroke $S = 82$ mm, clearance volume $V_c = 34.25$ cm^3, $\gamma = 1.4$.

Solution: $V_s = (\pi/4)(5.8)^2(8.2) = 216.65$ cm^3 → $V_1 = 216.65 + 34.25 = 250.90$ cm^3 → $r_v = 250.90 / 34.25 = 7.33$ → $\eta_{th} = 1 - (1 / 7.33^{0.4}) = 0.5492$ (**54.92%**).

EXAMPLE 2 — COMPLETE OTTO CYCLE NUMERICAL

Given: $T_1 = 300\ K$, $P_1 = 100\ kPa$, $Q_{in} = 1840\ kJ/kg$, $r_v = 8$.

- $v_1 = (0.287 \times 300)/100 = 0.861\ m^3/kg$, $v_2 = 0.861/8 = 0.1076\ m^3/kg$
- $P_2 = 100(8)^{1.4} = 1837.92\ kPa$, $T_2 = 300(8)^{0.4} = 689.22\ K$
- $1840 = 0.718(T_3 - 689.22) \Rightarrow T_3 = 3251.89\ K$, $P_3 = 1837.92 \times (3251.89/689.22) = 8671.71\ kPa$
- $P_4 = 8671.71(1/8)^{1.4} = 471.82\ kPa$, $T_4 = 3251.89(1/8)^{0.4} = 1415.47\ K$
- $q_{out} = 0.718(1415.47 - 300) = 800.91\ kJ/kg$, $\eta_{th} = 1 - (800.91/1840) = 0.5647$ (**56.47%**)
- $P_m = (1840 - 800.91) / (0.861 - 0.1076) = 1379.2\ kPa$

3. Air-Standard Diesel Cycle

Ideal cycle for CI engines with constant pressure heat addition (Process 2→3).

$$\text{Cut-off ratio: } r_c = V_3 / V_2$$

$$\eta_{th,\ Diesel} = 1 - (1 / r_v^{\gamma-1}) \left[(r_c^{\gamma} - 1) / (\gamma (r_c - 1)) \right]$$

DIESEL CYCLE NUMERICAL EXAMPLE

Given: $T_1 = 300\ K$, $P_1 = 100\ kPa$, $Q_{in} = 1840\ kJ/kg$, $r_v = 16$.

- $P_2 = P_3 = 100(16)^{1.4} = 4850.29\ kPa$, $T_2 = 300(16)^{0.4} = 909.43\ K$
- $1840 = 1.005(T_3 - 909.43) \Rightarrow T_3 = 2740.28\ K$ (Max Temp)
- $r_c = 2740.28 / 909.43 = 3.013 \rightarrow T_4 = 2740.28(0.1621/0.861)^{0.4} = 1405.10\ K$
- $q_{out} = 0.718(1405.10 - 300) = 793.46\ kJ/kg \rightarrow \eta_{th} = 1 - (793.46/1840) = 0.5688$ (**56.88%**)
- $P_m = (1840 - 793.46) / (0.861 - 0.0538) = 1296.51\ kPa$

4. Air-Standard Dual Cycle

Heat added in two stages: Process 2→3 (constant volume) and Process 3→4 (constant pressure).

$$\eta_{th,\ Dual} = 1 - (1 / r_v^{\gamma-1}) \left[(r_p r_c^{\gamma} - 1) / ((r_p - 1) + \gamma r_p (r_c - 1)) \right]$$

DUAL CYCLE NUMERICAL EXAMPLE

Given: $T_1 = 300\ K$, $P_1 = 100\ kPa$, $r_v = 16$, $P_{max} = 7500\ kPa$, $T_{max} = 2773\ K$.

- $P_2 = 4850\ kPa$, $T_2 = 909.43\ K \rightarrow T_3 = 909.43(7500/4850) = 1406.34\ K$
- $V_4/V_3 = 2773/1406.34 = 1.9718 \rightarrow T_5 = 2773 / (8.1152)^{0.4} = 1200.12\ K$
- $q_{in} = 0.718(1406.34\text{-}909.43) + 1.005(2773\text{-}1406.34) = 1730.27\ kJ/kg$
- $q_{out} = 0.718(1200.12 - 300) = 646.29\ kJ/kg \rightarrow \eta_{th} = 1 - (646.29/1730.27) = 0.6265$ (**62.65%**)
- $P_m = (1730.27 - 646.29)/(0.861 - 0.0538) = 1342.89\ kPa$

5. Stirling & Ericsson Cycles

Stirling (1815) consists of 2 isothermal and 2 constant-volume processes with regenerator. Ericsson (1853) consists of 2 isothermal and 2 constant-pressure processes with regenerator.

$$\eta_{th,\ Stirling} = \eta_{th,\ Ericsson} = 1 - (T_L / T_H) = \eta_{Carnot}$$

STIRLING CYCLE EXAMPLE

Given: $T_3 = T_4 = 460°C\ (673\ K)$, $P_1 = 12\ bar$, $P_3 = 2\ bar$, $r_v = 3$.

- $P_2 = 12 \times (1/3) = 4\ bar$, $P_4 = 2 \times 3 = 6\ bar$
- $T_1 = T_4(P_1/P_4) = 673 \times (12/6) = 1346\ K$
- $\eta_{th} = 1 - (673 / 1346) = 0.50 \rightarrow$ **50%**

ERICSSON CYCLE EXAMPLE

Given: $T_L = 45°C\ (318\ K)$, $T_H = 230°C\ (503\ K)$, expansion ratio $r_e = 2$, $R = 0.287\ kJ/kg{\cdot}K$.

- Work per kg: $w = R\ ln(r_e)(T_H - T_L) = 0.287 \times ln(2) \times (503 - 318) = 36.80\ kJ/kg$
- Efficiency: $\eta_{th} = 1 - (318/503) = 0.3678 \rightarrow$ **36.78%**

6. Vapour Power Cycles (Carnot & Rankine)

Carnot Vapour Cycle

Processes: 1→2 (Isothermal heating in boiler), 2→3 (Isentropic expansion in turbine), 3→4 (Isothermal cooling in condenser), 4→1 (Isentropic compression in compressor).

$$\eta_{th,\ Carnot} = 1 - (T_2 / T_1)$$

Disadvantages of Carnot Vapour Cycle: Low work ratio, severe difficulties in controlling partial condensation (process 3→4) and compressing liquid-vapour mixture, turbine blade erosion from high moisture content.

CARNOT VAPOUR EXAMPLE

Given: Dry saturated steam at $30\ bar$ ($T_1 = 233.8°C = 506.8\ K$) exiting at $0.02\ bar$ ($T_2 = 17.5°C = 290.5\ K$).

$\eta_{th} = 1 - (290.5 / 506.8) = 0.4268 \rightarrow$ **42.68%**

The Rankine Cycle

Modifies Carnot cycle by continuing condensation until steam is completely condensed to liquid water, utilizing a feedpump instead of a compressor.

First Law Derivation (Steady Flow Energy Equation):

- Boiler Heat Input: $Q_b = h_2 - h_1$
- Turbine Work Output: $W_t = h_2 - h_3$
- Condenser Heat Rejection: $Q_c = h_3 - h_4$

- Feed Pump Work Input: $W_p = h_1 - h_4 = v_f(P_1 - P_4)$
- Enthalpy derivation from $dh = du + d(pv)$ gives $dh = v\,dp$ for reversible adiabatic pump process: $\int dh = v \int dp \Rightarrow h_1 - h_4 = v_f(P_1 - P_4)$

$$\eta_{Rankine} = [\,(h_2 - h_3) - (h_1 - h_4)\,] / (h_2 - h_1) \approx (h_2 - h_3) / (h_2 - h_4)$$

Performance Criteria Definitions:

- **Efficiency Ratio:** $\text{Cycle Efficiency} / \text{Rankine Efficiency}$
- **Isentropic Turbine Efficiency:** $\eta_t = (h_2 - h_3') / (h_2 - h_3)$
- **Work Ratio:** $R_w = W_{net} / W_{gross} = (W_t - W_p) / W_t$
- **Specific Steam Consumption (SSC):** $SSC = 3600 / W_{net}$ [kg/kWh]

COMPREHENSIVE VAPOUR CYCLE COMPARISON PROBLEM

Limits: Boiler pressure = 42 bar, Condenser pressure = 0.035 bar.

Steam Table Data: At 42 bar: $T_{sat} = 253.2°C = 526.2$ K, $h_g = 2800$ kJ/kg, $s_g = 6.049$ kJ/kg·K.
At 0.035 bar: $T_{sat} = 26.7°C = 289.7$ K, $h_f = 112$ kJ/kg, $h_{fg} = 2438$ kJ/kg, $s_f = 0.391$ kJ/kg·K, $s_{fg} = 8.130$ kJ/kg·K.

(a) Carnot Vapour Cycle:

- $\eta_{Carnot} = 1 - (289.7 / 526.2) = 0.4304$ (**43.04%**)
- $s_3 = s_2 = 6.049 = 0.391 + x_3(8.130) \Rightarrow x_3 = 0.696$
- $h_3 = 112 + 0.696(2438) = 1808.84$ kJ/kg $\rightarrow W_t = 2800 - 1808.84 = 991.16$ kJ/kg
- $W_{net} = 0.4304 \times 1698 = 730.82$ kJ/kg
- $R_w = 730.82 / 991.16 = 0.7373$, $SSC = 3600 / 730.82 = 4.93$ kg/kWh

(b) Rankine Cycle (Dry Saturated):

- $W_p = 0.0010024 \times (42 - 0.035) \times 100 = 4.20$ kJ/kg
- $W_{net} = 991.16 - 4.20 = 986.96$ kJ/kg
- $\eta_R = 986.96 / (2800 - 112 - 4.2) = 0.3677$ (**36.8%**)
- $R_w = 986.96 / 991.16 = 0.9957$, $SSC = 3600 / 986.96 = 3.65$ kg/kWh

(c) Rankine Cycle with Isentropic Efficiency = 0.8:

- $W_t' = 0.8 \times 991.16 = 792.93$ kJ/kg $\rightarrow W_{net}' = 792.93 - 4.2 = 788.73$ kJ/kg
- $\eta_R' = 788.73 / 2683.8 = 0.294$ (**29.4%**)
- $R_w = (792.93 - 4.2) / 792.93 = 0.9947$, $SSC = 3600 / 788.73 = 4.56$ kg/kWh

7. Rankine Cycle with Superheat

Superheating steam above saturation temperature (typically to 480–540°C):

1. Increases thermal efficiency by elevating average temperature of heat addition.
2. Decreases Specific Steam Consumption.
3. Increases dryness fraction at turbine exhaust to prevent blade erosion.

SUPERHEAT NUMERICAL EXAMPLE

Given: Superheated steam at 42 bar, 500°C: $h_2 = 3442.6\ kJ/kg$, $s_2 = 7.066\ kJ/kg{\cdot}K$.

- $7.066 = 0.391 + x_3(8.130) \Rightarrow x_3 = 0.821$ (Dryness fraction increased from 0.696 to 0.821)
- $h_3 = 112 + 0.821(2438) = 2113.6\ kJ/kg \rightarrow W_t = 3442.6 - 2113.6 = 1329\ kJ/kg$
- $W_{net} = 1329 - 4.2 = 1324.8\ kJ/kg$
- $\eta_R = 1324.8 / (3442.6 - 112 - 4.2) = 0.3983$ (**39.8%** vs 36.8% dry saturated)
- $R_w = 1324.8 / 1329 = 0.997$, $SSC = 3600 / 1324.8 = 2.72\ kg/kWh$

8. Reheat Cycle

Main reason for reheating is to avoid wet steam condition at exit of turbine by increasing exhaust dryness fraction, while bringing an appreciable reduction in specific steam consumption.

$$\text{Turbine Work } W_t = W_{23} + W_{45} = (h_2 - h_3) + (h_4 - h_5)$$
$$\text{Pump Work } W_p = h_1 - h_6 = v_f(P_{boiler} - P_{condenser})$$
$$\eta_{Reheat} = [\ (h_2 - h_3) + (h_4 - h_5) - W_p\ ] / [\ (h_2 - h_1) + (h_4 - h_3)\ ]$$